Quantum samples, classical accuracy: a resource-efficient strategy for sample-based quantum diagonalization

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Sample-based quantum diagonalization (SQD) diagonalizes the Hamiltonian in determinants built from spin strings sampled on a quantum processor. It has been argued that such quantum-sampled configurations fall behind those chosen by classical selected configuration interaction. For strongly correlated molecules, SQD also faces practical problems. The product space of the sampled strings can exceed what one computer can diagonalize, the samples are noisy and tied to one orbital basis, and states built from broken-symmetry references are spin contaminated. We present an SQD strategy that addresses these problems with configuration recovery of the samples, selected configuration interaction restricted to the sampled strings, orbital optimization alternated with selection, new measurements in the optimized orbitals, and spin projection with a second-order correction. Every step after the measurement runs on one node with eight GPUs. We test the proposed method by comparing it with selected configuration interaction over all strings at the same number of determinants and with the same orbital optimization. With local unitary cluster Jastrow (LUCJ) circuits and 73 s of quantum processing unit time on the `ibm_kingston` processor, the spin-projected energy of $[Fe_2S_2(SCH_3)_4]^{2-}$ ends 0.19 mE_h above the exact (full configuration interaction) energy. SqDRIFT circuits with 128 s end 0.22 mE_h above it, and selection over all strings ends 0.18 mE_h above it. For $[Fe_4S_4(SCH_3)_4]^{2-}$ the sampled strings cover at most 4.5% of all spin strings. With 3×10^6 determinants the spin-projected energies lie 11.2 mE_h (LUCJ) and 11.7 mE_h (SqDRIFT) above a density matrix renormalization group (DMRG) estimate of the ground-state energy, whereas selection over all strings, started from the same determinant, ends 11.2 mE_h above it. At this number of determinants, the LUCJ samples therefore matched classical selection, and the SqDRIFT samples ended 0.5 mE_h above it.

I. INTRODUCTION

Sample-based quantum diagonalization (SQD) [1, 2] uses a quantum processing unit (QPU) to propose the electronic configurations in which the Hamiltonian is diagonalized. A circuit prepares an approximation to the ground state, and each measured bitstring gives one determinant, made of one spin-up and one spin-down string, each listing the occupied orbitals of its spin. The Hamiltonian is then diagonalized in the product space of the sampled spin-up and spin-down strings. For strongly correlated molecules four problems arise. The product space grows as the product of the numbers of sampled strings, and a recent SQD study of an iron–sulfur cluster diagonalized subspaces of up to 3×10^{10} determinants on a supercomputer [3]. Noise changes electron numbers, so the samples must be corrected [2]. The strings are sampled in one orbital basis, which is rarely the best one for the final state. And when the mean-field description breaks spin symmetry, the states built from the samples are spin contaminated.

Here we present an SQD workflow that addresses these problems and runs every step after the measurement on one GPU node. The workflow prepares circuits from broken-symmetry references and measures them in a common orbital basis, corrects the noisy samples by configuration recovery, and selects determinants inside the product space of the measured and recovered strings instead of diagonalizing it. It then alternates quasi-Newton orbital optimization with selection until the energy converges, and measures again in the optimized orbitals as long as determinants built from the sampled strings carry less than 95% of the second-order correction. It projects the final state onto the target total spin and adds a second-order correction. We use two kinds of circuits, a local unitary cluster Jastrow (LUCJ) circuit [4] for each reference, and SqDRIFT [5], which samples many shallow randomized time evolutions.

We apply the workflow to two iron–sulfur clusters, a class that includes the electron carriers of ferredoxins and the cofactors of nitrogenase. Their iron centers carry large local spins coupled antiferromagnetically, and the low-energy states are spin-coupled combinations of many near-degenerate configurations [6, 7]. Mean-field theory describes them only by breaking spin symmetry [8], and a mixed-valence cluster has several broken-symmetry references that differ in the placement of the extra electrons (valence isomers [9]). These clusters exercise every element of the workflow, and accurate reference energies are available for both.

A further question is whether quantum samples help at all. Recent studies argue that configurations sampled on

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a quantum processor fall behind classical selected configuration interaction [10, 11], in part because sampling keeps returning configurations already found [10]. Our main test addresses this claim. We compare the workflow with CIPSI (configuration interaction by perturbative selection made iteratively) [12] over all strings, at the same number of determinants and with the same orbital optimization. Section II describes the workflow. Section III tests it on [2Fe-2S], first in noiseless simulations against the exact energy and then on `ibm_kingston`, and Section IV applies it to the [4Fe-4S] cluster on the same processor. Section V discusses what the quantum processor and each step contribute, and what the workflow costs.

II. WORKFLOW

Figure 1 shows one round of the workflow. The optimized orbitals of a round define the measurement basis of the next. The first round uses a starting basis B_0 of split-localized natural orbitals [13, 14], computed from the active-space integrals alone. These are the natural orbitals of the lowest broken-symmetry unrestricted Hartree–Fock (UHF) solution, grouped by occupation and localized within each group by the Edmiston–Ruedenberg criterion [15].

A. Circuits

When the mean-field description breaks spin symmetry, the circuits start from broken-symmetry references, UHF solutions with definite local spin directions. Because references related by a global spin flip describe the same spin-coupled state, and spin projection (Sec. II E) generates the flipped configurations, we sample only one reference of each such pair. References that no spin operation relates, such as the valence isomers of a mixed-valence cluster [9], each get their own circuit, and their weights divide the shots among them (Sec. IV).

For each reference we build a one-layer LUCJ circuit [4] from the unrestricted coupled-cluster singles and doubles (UCCSD) amplitudes of its UHF determinant [16]. We add the opposite-spin Jastrow matrix to the same-spin matrix of each spin. This removes all gates between the spin-up and spin-down registers, so that the circuit state is a product of a spin-up and a spin-down state. For large active spaces the Jastrow and orbital-rotation layers act on a block of occupied and virtual orbitals chosen by a singular-value decomposition of the amplitudes (nine of each for [4Fe-4S]). A final orbital rotation takes each circuit to the common measurement basis B, so a change of B changes only this rotation.

The second circuit family, SqDRIFT [5], uses many shallow circuits. Each applies to a reference determinant a short time evolution built by random sampling of Hamiltonian terms (qDRIFT [17]) and is measured with 50–100 shots. The first round starts from the broken-symmetry references, later rounds from the 64 leading determinants of the state of the previous round, expressed in the new basis. Appendix B describes the sampling of terms and evolution times.

B. Sampled strings and configuration recovery

On hardware, where noise changes electron numbers, each round runs the configuration-recovery loop of Ref. [2] on its own samples, in its own basis. The first pass keeps the strings with the correct electron number. Each further pass uses the spin-resolved orbital occupations of the previous solution to change the occupations of every string with a wrong electron number, and repeats the selection of Sec. II C. The recovery loop ends when the energy changes by less than 0.1 mE_h, or after six passes including the first. The sampled strings of a round are all its strings with the correct electron number, measured or corrected. The string sets $\mathcal{S}_\uparrow$ and $\mathcal{S}_\downarrow$ of a round are the unions of the sampled strings of that round and of all earlier rounds.

C. Selection inside the sampled strings

Instead of diagonalizing in the full product space $\mathcal{S}_\uparrow \times \mathcal{S}_\downarrow$, we select determinants inside it with CIPSI [12] restricted to the sampled strings. Each selection starts from one determinant, which in hardware runs is formed by the most frequent sampled spin-up and spin-down strings. Each iteration evaluates the Epstein–Nesbet second-order contribution of the determinants of $\mathcal{S}_\uparrow \times \mathcal{S}_\downarrow$ connected to the current space by H and adds the largest ones, doubling the space. Determinants from different references enter the same eigenvector, so the Hamiltonian determines how the references mix. On hardware the corrected string sets can contain nearly all spin strings, and the selection stops at a preset size D. In the noiseless simulations it stops at D for SqDRIFT and runs until no candidates remain for LUCJ.

D. Orbital optimization and the next measurement

After each selection the determinant set $\mathcal{V}$ is kept fixed, and we optimize the orbitals for it by quasi-Newton (BFGS) steps that minimize

$$E(\kappa) = \min_{\mathbf{c}} \frac{\mathbf{c}^\mathsf{T} H_\mathcal{V}(\kappa)\, \mathbf{c}}{\mathbf{c}^\mathsf{T}\mathbf{c}}, \tag{1}$$

the lowest eigenvalue in $\mathcal{V}$ with the integrals rotated by $U = e^\kappa$, where κ is a real antisymmetric matrix over the active orbitals and $\mathbf{c}$ the vector of configuration interaction (CI) coefficients. The gradient follows from the reduced density matrices of the eigenvector [18], as in orbital-optimized selected CI [19]. The eigenproblem in $\mathcal{V}$ is

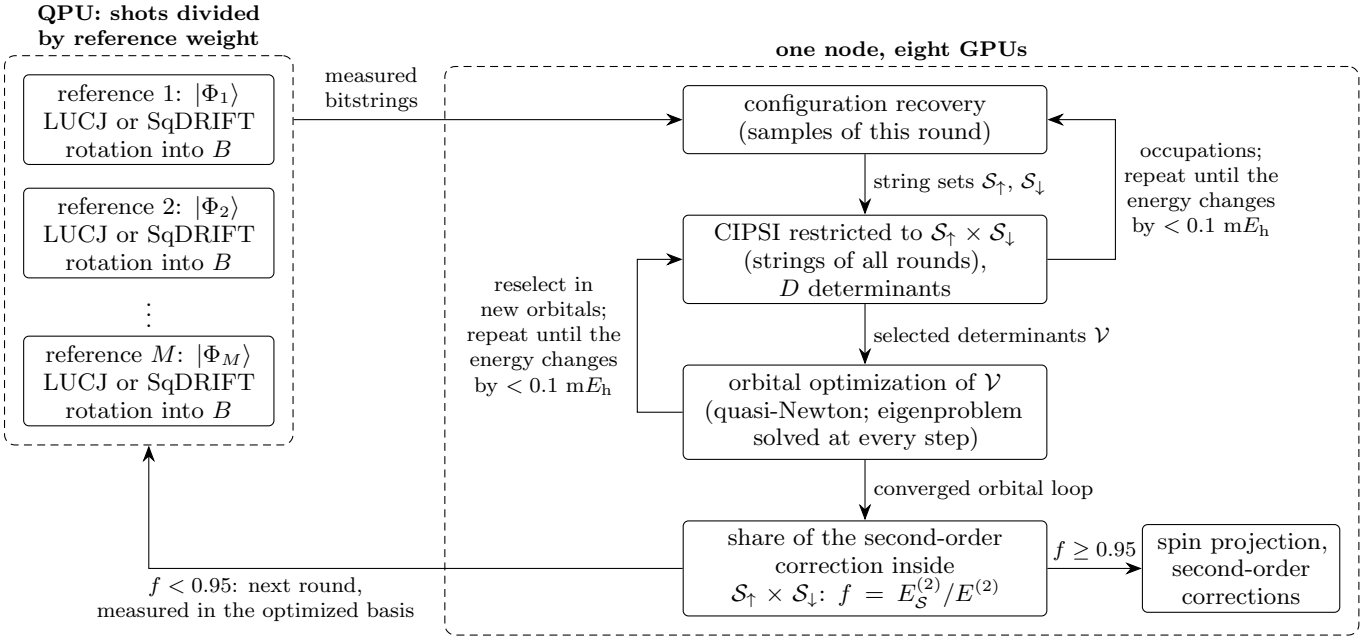


FIG. 1. One round of the workflow. Each broken-symmetry reference $|\Phi_m\rangle$ has a circuit (one LUCJ circuit, or many SqDRIFT circuits) whose last layer rotates the orbitals into the measurement basis B (B_0 in round 1). On hardware, configuration recovery and the selection alternate until the energy changes by less than 0.1 mE_h. The recovered strings are added to those of earlier rounds to form the sets $\mathcal{S}_\uparrow$ (spin up) and $\mathcal{S}_\downarrow$ (spin down). Selection and orbital optimization of the selected determinants $\mathcal{V}$ then alternate (the orbital loop) until an alternation lowers the energy by less than 0.1 mE_h. If the determinants of $\mathcal{S}_\uparrow \times \mathcal{S}_\downarrow$ outside $\mathcal{V}$ carry less than 95% of the second-order correction ($f < 0.95$), the circuits are measured again in the optimized basis. Otherwise, or after the third round, the state is projected onto $S = 0$ and corrected to second order.

solved again at every step, so energy and gradient always use the optimal CI coefficients for the current orbitals. The optimization stops after three consecutive steps that each lower the energy by less than $10^{-5}\,E_\mathrm{h}$.

In the optimized orbitals other determinants of $\mathcal{S}_\uparrow \times \mathcal{S}_\downarrow$ can have larger second-order contributions than some of those in $\mathcal{V}$, and the optimal orbitals depend in turn on the determinant set. We therefore select again in the optimized orbitals, from the same starting determinant and string sets, and optimize the orbitals for the new set. Selection and orbital optimization alternate in this way (the orbital loop) until an alternation lowers the energy by less than 0.1 mE_h. Both steps aim to lower the same quantity, the lowest eigenvalue over sets of at most D determinants of $\mathcal{S}_\uparrow \times \mathcal{S}_\downarrow$ and over orbital rotations. The strings sampled in the measurement basis are reused as occupation patterns of the rotated orbitals, and Appendix D tests this reuse.

A new measurement in the optimized orbitals can add strings that the earlier ones lack, because the string probabilities of the circuit state in the rotated basis depend on the phases of its amplitudes, which a measurement in the earlier basis does not record. Whether it is needed follows from the Epstein–Nesbet second-order correction of the selected state $|\psi\rangle = \sum_{b\in\mathcal{V}} c_b|b\rangle$, with variational energy E,

$$E^{(2)} = \sum_{a\notin\mathcal{V}} \frac{|\langle a|H|\psi\rangle|^2}{E - H_{aa}}, \tag{2}$$

summed over the determinants a connected to $\mathcal{V}$ by H. Restricting a to $\mathcal{S}_\uparrow \times \mathcal{S}_\downarrow$ gives $E^{(2)}_{\mathcal{S}}$, and the share of the second-order correction carried by determinants built from the sampled strings is

$$f = \frac{E^{(2)}_{\mathcal{S}}}{E^{(2)}}. \tag{3}$$

If $f \geq 0.95$ the workflow stops, and otherwise the circuits are measured again in the optimized basis. We run at most three rounds, a limit set by the available QPU time. When the selection exhausts $\mathcal{S}_\uparrow \times \mathcal{S}_\downarrow$, as for noiseless LUCJ samples, $f = 0$ and every round adds a measurement.

E. Spin projection and second-order corrections

We project the final state onto total spin $S = 0$ with the Löwdin projector P_0 [20]. The spin-projected energy $E_\mathrm{P} = \langle\psi|HP_0|\psi\rangle/\langle\psi|P_0|\psi\rangle$ is variational because P_0 commutes with H.

The final state is corrected to second order by $E^{(2)}$ of

Eq. (2), and the spin-projected state by

$$E_{\mathrm{P}}^{(2)} = \frac{1}{w} \sum_{K \notin \mathcal{K}(\mathcal{V})} \frac{\langle \sigma_K | P_0 | \sigma_K \rangle}{E_{\mathrm{P}} - \bar{E}_K}, \tag{4}$$

where K runs over the spatial configurations (orbital occupations 0, 1 or 2) outside $\mathcal{K}(\mathcal{V})$, the configurations of the determinants in $\mathcal{V}$. Here σ_K is $H|\psi\rangle$ restricted to the determinants of K, $w = \langle\psi|P_0|\psi\rangle$ is the singlet weight of $|\psi\rangle$, and $\bar{E}_K$ is the diagonal energy of K averaged over its spin couplings for $S = 0$. Appendix A describes the screening of small contributions and compares the projection and all corrections with evaluations in the full CI space of [2Fe-2S].

To estimate the limit of large D, we repeat the selection of the final round at $D/4$ and $D/2$, in the final orbitals and with the final string sets, and extrapolate the spin-projected energies of the three states linearly in $E_{\mathrm{P}}^{(2)}$ to $E_{\mathrm{P}}^{(2)} = 0$.

We report each energy X as its error in mE_{h}, $\Delta X = X - E_{\mathrm{FCI}}$ for [2Fe-2S] and $\Delta X = X - E_{\mathrm{DMRG}}$ for [4Fe-4S]. Here E_{FCI} is the exact (full configuration interaction, FCI) energy (Sec. III) and E_{DMRG} a density matrix renormalization group (DMRG) estimate of the ground-state energy (Sec. IV). For a state we give ΔE, $\Delta(E + E^{(2)})$, ΔE_{P} and $\Delta(E_{\mathrm{P}} + E_{\mathrm{P}}^{(2)})$, the errors of the variational energy, of the variational energy with its second-order correction, of the spin-projected energy and of the spin-projected energy with its second-order correction.

F. Implementation

All steps after the measurement run on one node with eight NVIDIA V100 GPUs, using the distributed operator of Fulqrum [21] and the PRIMME eigensolver [22]. Configuration recovery is a GPU port of the procedure in qiskit-addon-sqd [23]. UHF solutions and UCCSD amplitudes are computed with PySCF [16], and the circuits are built with ffsim [24] and Qiskit [25]. Appendix A describes the GPU kernels and their validation, Appendix B the circuits.

III. [2Fe-2S] CLUSTER

Setup. $[Fe_2S_2(SCH_3)_4]^{2-}$ in the CAS(30e,20o) active space of Li and Chan [26] has 15,504 strings per spin and 2.4×10^8 determinants. Its exact energy, $E_{\mathrm{FCI}} = -116.605609\,E_{\mathrm{h}}$, obtained by diagonalization in the full space, agrees with the DMRG energy of Ref. [26]. Because the cluster has a single valence isomer, each round measures one LUCJ circuit or one set of SqDRIFT circuits. In the noiseless simulations the LUCJ circuit is sampled from its spin-up and spin-down string distributions independently, which is exact because its state is a product of a spin-up and a spin-down state (Sec. II A). SqDRIFT circuits are simulated exactly in the determinant basis. Each noiseless calculation runs three rounds and is repeated with three random seeds. On `ibm_kingston` (IBM Heron r2, 156 qubits) the LUCJ circuit has 1,092 controlled-Z (CZ) gates at two-qubit depth 80, and the SqDRIFT circuits have 57–69 CZ gates on average, depending on the round. A hardware round uses 10^6 LUCJ shots or 6,000 SqDRIFT circuits of 50 shots, and the selection stops at $D = 1.1 \times 10^6$. As the classical baseline we run CIPSI over all strings with the same orbital loop, the same D and the same starting determinant as the LUCJ run.

Noiseless simulation. With SqDRIFT circuits and the selection stopped at $D = 10^6$, the orbital loop of the first round lowers ΔE from 3.91 to 2.46 mE_{h} in two or three alternations (Table I). The combined string sets grow from about 3,400 to 4,900 and 5,800 strings per spin over the three rounds, and f grows from 0.07 to 0.31, below the threshold in every round. After three rounds ΔE_{P} is 0.21 mE_{h}, against 0.20 mE_{h} for CIPSI over all strings, while ΔE is 1.96 against 1.55 mE_{h}. Appendix C shows that the orbital loop and the combination of the strings of all rounds each lower ΔE_{P}.

The LUCJ circuit puts most of its weight on few strings. Its 10^6 shots in B_0 give about 430 distinct strings per spin, and the selection exhausts $\mathcal{S}_\uparrow \times \mathcal{S}_\downarrow$. Because no other determinants can be selected, the first orbital loop ends where a single orbital optimization does (ΔE from 42.6 to 8.9 mE_{h}). Each further measurement adds strings and determinants. After three rounds ΔE_{P} is 0.76 mE_{h} at 10^6 shots per round and 0.45 mE_{h} at 10^7, and it is within chemical accuracy (1.6 mE_{h}) for every seed from the second round on.

On `ibm_kingston`. With the LUCJ circuit, 20–25% of the measured spin strings have the correct electron number. These include about 12,000 distinct strings per spin, against about 430 in noiseless samples of the same size. Since 500 strings carry 99.96% of the noiseless distribution, most of the additional strings come from noise. After recovery the string sets contain 99.4–99.6% of all spin strings, the orbital loop ends at $f = 1.00$, and the workflow stops after one round. The run uses 73 s of QPU time and ends at $\Delta E = 1.38$ and $\Delta E_{\mathrm{P}} = 0.19$ mE_{h}, against 1.35 and 0.18 mE_{h} for CIPSI over all strings (Table I). Noise and recovery thus give the hardware samples nearly the whole string space, which the noiseless LUCJ samples lack.

With the shallower SqDRIFT circuits, 61–65% of the spin strings have the correct electron number, and the recovered sets of the first round hold 22% of all spin strings. At $f = 0.06$ after the first round the circuits are measured again in the optimized orbitals. A third round follows at $f = 0.14$, after which the strings of the three rounds cover 34% of all spin strings. After 128 s of QPU time, the workflow ends at $\Delta E = 2.01$ mE_{h} and $\Delta E_{\mathrm{P}} = 0.22$ mE_{h}, against 1.35 and 0.18 mE_{h} for CIPSI over all strings. At $f = 0.22$ the criterion $f \geq 0.95$ calls for rounds beyond the three that we ran.

The final states are far from singlets ($w = 0.31$–0.42), and spin projection removes most of their error (Table I).

TABLE I. [2Fe-2S], errors $\Delta X = X - E_{\mathrm{FCI}}$ in mE_{h}. ΔE is the variational energy after the first selection (after recovery on hardware) and after the orbital loop of each round. $\Delta(E + E^{(2)})$ (unscreened), ΔE_{P} and $\Delta(E_{\mathrm{P}} + E_{\mathrm{P}}^{(2)})$ refer to the state of the last round, and extrap. is ΔE_{P} extrapolated to $E_{\mathrm{P}}^{(2)} = 0$ (Sec. II E), computed for hardware only. f is the share of the second-order correction inside $\mathcal{S}_\uparrow \times \mathcal{S}_\downarrow$ in the last round, Eq. (3). Shots are per round, and SqDRIFT circuits have 50 shots each. Noiseless rows are means over three seeds, with D fixed for SqDRIFT and set by the exhausted string sets for LUCJ (mean of the last round). CIPSI runs over all strings with one orbital loop. QPU time is the billed QPU time of all rounds.

	ΔE										
	first	after round									
	selection	1	2	3	$\Delta(E + E^{(2)})$	ΔE_{P}	$\Delta(E_{\mathrm{P}} + E_{\mathrm{P}}^{(2)})$	extrap.	f	D	QPU time
noiseless											
SqDRIFT, 10^6 shots	3.91	2.46	2.08	1.96	1.71	0.21	0.18	–	0.31	1.0×10^6	–
LUCJ, 10^6 shots	42.6	8.93	3.95	3.71	3.12	0.76	0.70	–	0	1.8×10^6	–
LUCJ, 10^7 shots	20.4	5.73	3.27	2.95	2.53	0.45	0.44	–	0	5.8×10^6	–
CIPSI, all strings	2.57	1.55	–	–	1.30	0.20	0.17	–	–	1.0×10^6	–
`ibm_kingston`											
LUCJ, 10^6 shots	2.43	1.38	–	–	1.13	0.19	0.17	0.11	1.00	1.1×10^6	73 s
SqDRIFT, 3×10^5 shots	4.16	2.37	2.13	2.01	1.75	0.22	0.20	0.15	0.22	1.1×10^6	128 s
CIPSI, all strings	2.43	1.35	–	–	1.11	0.18	0.16	0.11	–	1.1×10^6	–

Extrapolated to $E_{\mathrm{P}}^{(2)} = 0$, ΔE_{P} is 0.11 (LUCJ), 0.15 (SqDRIFT) and 0.11 mE_{h} (CIPSI over all strings).

IV. [4Fe-4S] CLUSTER

Setup. $[Fe_4S_4(SCH_3)_4]^{2-}$ in the CAS(54e,36o) active space of Li and Chan [26] has 9.4×10^7 strings per spin. As reference we use $E_{\mathrm{DMRG}} = -327.248(2)\ E_{\mathrm{h}}$, the DMRG-based estimate of the ground-state energy of Ref. [27]. For this mixed-valence cluster we take from a small selected CI wave function (3.45×10^5 determinants) the eight charge distributions of largest weight, each a placement of the extra electrons on the four iron sites, and start a broken-symmetry UHF calculation from the leading determinant of each. Removing one duplicate solution and one calculation without converged coupled-cluster amplitudes leaves six valence isomers, whose weights in the selected CI wave function divide the shots among them. A noiseless test in B_0 supports the use of six isomers (Appendix B). On `ibm_kingston` the six valence-isomer LUCJ circuits have 2,872–2,897 CZ gates at two-qubit depth 104–108. A round takes 5×10^6 LUCJ shots or 32,000 SqDRIFT circuits (100 CZ gates on average) of 100 shots at comparable QPU time, and the selection stops at $D = 3 \times 10^6$. As the classical baseline we run CIPSI over all strings with the same orbital loop and the same D, starting in B_0 from the same determinant as the LUCJ run. To see how much the start matters, we also start it from the leading determinants of the six valence isomers and from the lowest UHF determinant and its global spin flip.

Results. Here the sampled strings restrict the selection. Noise affects these deeper circuits more than those of [2Fe-2S]. Of the LUCJ spin strings, 4.5–5.9% have the correct electron number, and 0.24–0.30% of the shots have it in both spin strings. These spin strings include about 2×10^5 distinct strings per spin, and after recovery the string sets of one round hold 2.1–2.5 $\times 10^6$ strings per spin, 2–3% of all spin strings.

In the first LUCJ round (Table II) recovery lowers ΔE from 59.6 to 38.4 mE_{h}, the first orbital optimization to 25.9 mE_{h}, and the full orbital loop of nine alternations to 15.2 mE_{h}. The orbital loop rotates the orbitals far from B_0, and the 20 leading determinants keep on average 4% of their weight on their occupation patterns in B_0 (the quantity p of Appendix D). Determinants built from the strings sampled in B_0 still carry 81% of the second-order correction. Since $f = 0.81 < 0.95$, the circuits are measured again in the optimized orbitals. Selection over the new strings gives $\Delta E = 14.9$ mE_{h} after recovery, below the 15.2 mE_{h} at the end of the first round. Combined with those of the first round, they form string sets of 4.2×10^6 spin-up and 3.8×10^6 spin-down strings (4.5 and 4.0% of all spin strings). The orbital loop converges in one alternation, and $f = 0.95$ stops the workflow after 10^7 shots and 13 min of QPU time. The final state has $\Delta E_{\mathrm{P}} = 11.2$ mE_{h} and $\Delta(E_{\mathrm{P}} + E_{\mathrm{P}}^{(2)}) = 8.7$ mE_{h}. Although recovery adds most of the strings, 77% of the determinants of this state, carrying 99.5% of the weight, have both strings measured.

For the SqDRIFT circuits, 44–45% of the spin strings have the correct electron number, but the samples contain fewer distinct strings, 4.7–7.1$\times 10^4$ per spin in a round and 1.0–1.5 $\times 10^5$ after recovery (0.1–0.2% of all spin strings). In the first round the first orbital optimization lowers ΔE from 50.9 to 29.9 mE_{h}, and the full orbital loop of five alternations to 16.8 mE_{h}. Two more rounds follow at $f = 0.47$ and 0.66, each lowering ΔE further. The strings of the three rounds, 2.5×10^5 spin-up and 2.6×10^5 spin-

TABLE II. [4Fe-4S] on `ibm_kingston` at $D = 3\times10^6$, round by round, as errors $\Delta X = X - E_{\mathrm{DMRG}}$ in mE_h. ΔE is the variational energy after selection over the round's measured strings with the correct electron number (meas.), after its recovery loop (rec.), after selection over the strings of all rounds so far (comb.) and after the orbital loop. $\Delta(E + E^{(2)})$ (screening $\epsilon = 10^{-7}\,E_\mathrm{h}$), ΔE_P and $\Delta(E_\mathrm{P} + E_\mathrm{P}^{(2)})$ refer to the state at the end of the round, and extrap. is ΔE_P of the final state extrapolated to $E_\mathrm{P}^{(2)} = 0$ (Sec. II E). f is the share of the second-order correction inside $\mathcal{S}_\uparrow \times \mathcal{S}_\downarrow$, Eq. (3), and QPU the billed QPU time up to the end of the round. LUCJ uses six isomers and 5×10^6 shots per round, SqDRIFT 32,000 circuits of 100 shots per round. CIPSI runs over all strings, with a first selection in B_0 and one orbital loop, started from the LUCJ starting determinant, from the six isomer determinants or from the lowest UHF determinant and its spin flip.

		ΔE									
	round	meas.	rec.	comb.	orbital loop	$\Delta(E + E^{(2)})$	ΔE_P	$\Delta(E_\mathrm{P} + E_\mathrm{P}^{(2)})$	extrap.	f	QPU
LUCJ	1	59.6	38.4	–	15.2	10.8	11.8	9.1	–	0.81	380 s
	2	21.3	14.9	14.7	14.7	10.6	11.2	8.7	4.8	0.95	761 s
SqDRIFT	1	58.3	50.9	–	16.8	11.6	13.4	10.2	–	0.47	331 s
	2	17.7	16.2	15.6	15.6	11.1	12.2	9.4	–	0.66	660 s
	3	16.9	15.7	15.2	15.1	10.9	11.7	9.1	4.9	0.78	988 s
CIPSI, LUCJ start	–		37.9		14.6	10.6	11.2	8.7	–	–	–
CIPSI, isomer starts	–		41.0		14.6	10.5	11.1	8.6	–	–	–
CIPSI, UHF start	–		31.4		18.9	13.7	14.1	11.1	4.9	–	–

down strings, cover 0.27% of all spin strings. The final state, after 16 min of QPU time against 13 min for LUCJ, has $\Delta E_\mathrm{P} = 11.7$ mE_h and $\Delta(E_\mathrm{P} + E_\mathrm{P}^{(2)}) = 9.1$ mE_h.

From the same starting determinant, CIPSI over all strings ends where the LUCJ run does. ΔE_P is 11.2 mE_h for both, and the other three energies agree within 0.1 mE_h (Table II). The SqDRIFT run ends at 11.7 mE_h. Started from the six isomer determinants, CIPSI gives the same result as from the LUCJ start, within 0.1 mE_h. From the lowest UHF determinant and its spin flip, however, its orbital loop ends higher, at $\Delta E_\mathrm{P} = 14.1$ mE_h, so the selection and the orbital loop depend on where they start. Linear extrapolation in $E_\mathrm{P}^{(2)}$ through $D/4$, $D/2$ and D gives ΔE_P of 4.8 (LUCJ), 4.9 (SqDRIFT) and 4.9 mE_h (CIPSI from the UHF start). For comparison, with a single broken-symmetry reference, the same Hamiltonian and no orbital optimization, Ref. [3] reports, with 3×10^{10} determinants, a variational energy 12.92 mE_h above E_{DMRG}, and an extrapolated energy 3.5 mE_h above it.

TABLE III. Cost of the LUCJ and SqDRIFT hardware rounds. Recovery passes include the first, and alternations are those of the orbital loop.

	[2Fe-2S]	[4Fe-4S]
selection size D	1.1×10^6	3×10^6
recovery passes per round	3	3–6
recovery and selection, per pass	15–17 s	45–70 s
alternations, round 1 / later	2–6 / 1	5–9 / 1
one alternation	40–110 s	2.5–9 min
projection, corrections and f	1–2 min	7–9 min
QPU time per round	33–73 s	328–381 s

V. DISCUSSION AND CONCLUSIONS

Quantum samples against classical selection. Our comparison with CIPSI over all strings tests the claim that configurations sampled on a quantum processor fall behind classical selected configuration interaction [10, 11]. On [2Fe-2S], noise and recovery give the LUCJ run nearly the whole string space, and it reproduces CIPSI over all strings (ΔE_P of 0.19 against 0.18 mE_h). The SqDRIFT run, whose strings cover a third of the string space, ends at 0.22 mE_h. On [4Fe-4S], the string sets cover at most 4.5% of all strings, so the samples decide which determinants can be selected. Even so, from the same starting determinant the LUCJ run matches CIPSI over all strings within 0.1 mE_h in every energy we report, and the SqDRIFT run ends 0.5 mE_h above it (11.7 against 11.2 mE_h in ΔE_P). On [4Fe-4S] at the D used here, selection over the quantum samples was therefore as accurate as selection over all strings for LUCJ and nearly so for SqDRIFT. The starting determinant of the selection changes the result more than the restriction to the samples does. Both circuit families can still be sampled classically at these sizes, so the hardware runs test how the workflow handles noisy samples.

What each step contributes. Configuration recovery makes the noisy samples usable, but most of the weight of the final [4Fe-4S] LUCJ state lies on measured strings. On [4Fe-4S], the orbital loop lowers the energy well beyond a single orbital optimization, because an optimization of a fixed determinant set cannot replace determinants that the new orbitals make less important, whereas the

next selection can. A new measurement in the optimized orbitals adds strings that the first one lacks, and the share f decides whether to measure again. It stops a run once determinants built from the sampled strings carry the second-order correction, as for LUCJ, and asks for further rounds while they do not, as for SqDRIFT. Because the selected states remain far from singlets, spin projection removes most of the remaining error on [2Fe-2S] and about a quarter of it on [4Fe-4S].

Cost. In both clusters SQD did not need the full product space of the sampled strings. On [4Fe-4S] the combined LUCJ string sets span about 1.6×10^{13} determinants, and the selection keeps 3×10^6 of them. On the GPUs, a round of recovery, selection, orbital optimization, projection and corrections takes 2–6 min for [2Fe-2S] and 10–43 min for [4Fe-4S] on one node (Table III). Most of the total time is spent in the orbital loop of the first round.

Limitations. Because each hardware configuration was run once, we cannot tell whether the LUCJ and SqDRIFT results differ. The selection size was fixed in advance rather than converged, the orbital loop and the decision to measure again used preset thresholds (0.1 mE_h per alternation, $f \geq 0.95$), and the SqDRIFT runs stopped after three rounds with $f < 0.95$. The valence isomers of [4Fe-4S] and their weights come from a classical selected CI wave function. The comparison with CIPSI uses one D per cluster. The extrapolation is tested only on [2Fe-2S], where it ends 0.11–0.15 mE_h above the exact energy and lies less than 0.1 mE_h below ΔE_P at D. For [4Fe-4S] it moves ΔE_P from 11.2–14.1 to 4.8–4.9 mE_h.

None of the elements of the workflow is specific to iron–sulfur clusters. It needs only an active-space Hamiltonian, references for the circuits, and circuits whose last layer rotates into the measurement basis. Repeated hardware runs, a converged selection size and circuits that cannot be simulated classically are needed to establish what the quantum processor adds beyond the results reported here.

Appendix A: GPU implementation and validation

Each of the eight V100 GPUs (32 GB) runs one MPI rank. Every step was validated on [2Fe-2S] and [4Fe-4S] test cases.

Configuration recovery. The recovery is a GPU port of the procedure of qiskit-addon-sqd [23]. For each distinct bitstring whose spin-up or spin-down string has the wrong electron number, it switches orbitals of that string on or off one at a time, with probabilities set by the orbital occupations of the previous solution. The orbital weights agree with those of qiskit-addon-sqd to 4×10^{-16}. For three test bitstrings, 10^6 GPU corrections reproduce the exactly enumerated distribution of corrected strings. The total variation distance is 0.0003–0.0020, below the bound $\frac{1}{2}\sqrt{m/n}$ = 0.003–0.008 on its expected value for $n = 10^6$ samples over m = 28–280 outcomes, and no weight falls outside the support. One recovery pass on 2×10^7 [4Fe-4S] shots of a separate LUCJ experiment on `ibm_kingston` (1.9×10^7 bitstrings to correct) takes 0.5 s on the GPUs, against about 950 s for qiskit-addon-sqd on 48 CPU processes. It gives string sets of the same size (3.99×10^6 and 1.53×10^6 strings), and their overlap with the CPU sets equals the overlap between two GPU runs with different seeds.

Hamiltonian and eigensolver. The Hamiltonian in the selected space is the distributed operator of Fulqrum [21], and its lowest eigenpair is computed with PRIMME [22], warm-started from the previous eigenvector. Two steps of the operator construction, the generation and the merging of the lists of connected strings, run on the GPUs. On [4Fe-4S] test cases with up to 3.7×10^8 list entries their output is bitwise identical to that of Fulqrum's CPU implementation and is produced about three times faster.

Selection. In each iteration every rank computes, from its share of the current eigenvector, the contributions $H_{ab}c_b$ to the residuals $r_a = \sum_b H_{ab}c_b$ of the determinants a that lie outside the current space and have their spin-up and spin-down strings in $\mathcal{S}_\uparrow$ and $\mathcal{S}_\downarrow$. Contributions below a screening threshold (initially $10^{-4}\,E_\mathrm{h}$, then a quarter of the smallest selected $|r_a|$, at least $10^{-7}\,E_\mathrm{h}$) are skipped. The contributions are summed per determinant on the rank that owns it, and the Epstein–Nesbet contributions $e_a = r_a^2/(E - H_{aa})$ are compared across MPI ranks by a bisection on a common threshold. Compared with Fulqrum's CPU (OpenMP) residual, the GPU residual finds identical candidate sets (7.3×10^4 for a [2Fe-2S] space of 5,000 determinants, 9.1×10^6 for a [4Fe-4S] space of 2×10^5 determinants), with residuals that agree to $2 \times 10^{-18}\,E_\mathrm{h}$ and diagonal elements that agree to $5 \times 10^{-12}\,E_\mathrm{h}$.

Density matrices and orbital gradient. The one- and two-particle reduced density matrices are computed on the GPUs with Fulqrum. At every step of the orbital optimization we check that the energy evaluated from them agrees with the eigenvalue to $10^{-6}\,E_\mathrm{h}$. For a [4Fe-4S] space of 2.7×10^6 determinants the difference is $6 \times 10^{-14}\,E_\mathrm{h}$. The analytic orbital gradient, which uses the derivative of the matrix exponential, agrees with finite differences to a relative 3×10^{-7} in the same space.

Spin projection. With $\mathcal{D}_K$ the determinants of spatial configuration K, the two matrix elements of Sec. II E are

$$\langle\psi|P_0|\psi\rangle = \sum_K \sum_{b,b'\in\mathcal{D}_K} c_b c_{b'} \langle b|P_0|b'\rangle, \tag{A1}$$

$$\langle\psi|HP_0|\psi\rangle = \sum_b c_b \sum_a H_{ab}\,(P_0\psi)_a, \tag{A2}$$

where $\langle b|P_0|b'\rangle$ is a Sanibel coefficient [28], determined up to a sign by the number of open shells of K and the number of them with the same spin in b and b'. The index a runs over b and its single and double excitations whose configuration occurs in the space, and $(P_0\psi)_a = \sum_{b'\in\mathcal{D}_{K(a)}} \langle a|P_0|b'\rangle c_{b'}$. We validated the evaluator in three steps. (i) For five selected CI states of [2Fe-2S] with 10^4 to 8×10^6 determinants, a CPU implementation of the

same formulas reproduces to $10^{-7}\,E_\mathrm{h}$ the spin-projected energies obtained exactly by applying $P_0 = \prod_{S'=1}^{5}[S^2 - S'(S'+1)]/[-S'(S'+1)]$ in the full CI space (2.4×10^8 determinants) with PySCF [16]. (ii) The GPU and CPU implementations agree to $6\times10^{-14}\,E_\mathrm{h}$ for a [2Fe-2S] state of 2×10^5 determinants and to $3\times10^{-12}\,E_\mathrm{h}$ for a [4Fe-4S] state of 2.7×10^6 determinants (1.9×10^6 configurations, up to 18 open shells), where the GPU evaluation takes 7.9 s against 593 s on 48 CPU cores. (iii) Replacing $(P_0\psi)_a$ by c_a reproduces the variational energy to $10^{-13}\,E_\mathrm{h}$.

Second-order corrections. Contributions $H_{ab}c_b$ with $|H_{ab}c_b| < \epsilon$ are skipped, with $\epsilon = 10^{-7}\,E_\mathrm{h}$ for [4Fe-4S] ($5\times10^{-7}\,E_\mathrm{h}$ for f) and $\epsilon = 0$ (no screening) for [2Fe-2S]. For [4Fe-4S], tightening ϵ from 5×10^{-7} to $10^{-7}\,E_\mathrm{h}$ changes the second-order corrections by at most 0.07 mE_h. The corrections use the same residual kernel as the selection, over all strings or, for $E_\mathcal{S}^{(2)}$, over $\mathcal{S}_\uparrow \times \mathcal{S}_\downarrow$. The contributions are summed on the rank that owns their spatial configuration, so that all determinants of a configuration meet on one rank. The correction of the spin-projected state applies P_0 within each configuration through the Sanibel coefficients in closed form. For [2Fe-2S] selected states of 62,500 and 10^6 determinants, the GPU values of $E^{(2)}$ and $E_\mathrm{P}^{(2)}$ without screening agree to 10^{-6} mE_h with evaluations in the full CI space that apply P_0 to the full-space vector with PySCF [16]. For the first state $E_\mathcal{S}^{(2)}$ agrees to the same precision.

Appendix B: Circuit details

Valence isomers. In one noiseless round in B_0 without orbital optimization (5×10^6 shots, $D = 3\times10^6$), the six isomers give $\Delta E_\mathrm{P} = 41.7$ mE_h, against 44.9 mE_h for the isomer of largest weight alone. With the same number of shots, the six circuits sample 45,670 spin-up and 32,063 spin-down strings, against 38,458 and 26,348 for that isomer alone.

SqDRIFT. The off-diagonal Hamiltonian is grouped as $H_\mathrm{off} = \sum_g \lambda_g G_g$, where each Hermitian group operator G_g contains an excitation, its Hermitian conjugate and the terms that share its coefficient through the symmetries of the integrals, each with coefficient ±1, and λ_g is the mean coefficient magnitude of the group. A circuit draws N groups ($N = 5$ and 10 in alternate circuits) with probabilities proportional to $a_g\lambda_g$, where $a_g = 1$ for groups that act on its reference determinant and $a_g = 0.05$ otherwise. It keeps a drawn group only if the group acts on at least one spin orbital that can be occupied and one that can be empty in the state prepared so far (the occupied and empty orbitals of the reference, plus all orbitals acted on by earlier groups). The circuit then applies $\prod_l e^{-i\delta G_{g_l}}$ with $\delta = \lambda_\mathrm{eff}t/N$ and $\lambda_\mathrm{eff} = \sum_g a_g\lambda_g$. The time t is drawn from a Gaussian with a scale chosen in each round so that on average 45% of the probability lies outside the reference determinant.

Hardware circuits. For [2Fe-2S] an exact simulation of the hardware circuit reproduces the noiseless string distribution of each register to a total variation distance of 3×10^{-12}. All jobs used the SamplerV2 primitive with XX dynamical decoupling, measurement twirling and a repetition delay of 50 μs.

TABLE IV. [2Fe-2S], noiseless SqDRIFT with 10^6 shots per round and $D = 10^6$. The entries are ΔE_P (ΔE in parentheses) in mE_h, means over three seeds, after two and three rounds. The strings are those of each round alone or of all rounds combined, and one opt. denotes a single orbital optimization per round.

	2 rounds	3 rounds
one opt., strings of each round	0.31 (2.33)	–
orbital loop, strings of each round	0.28 (2.26)	0.29 (2.30)
orbital loop, strings of all rounds	0.23 (2.08)	0.21 (1.96)
CIPSI, all strings, orbital loop	0.20 (1.55)	

Appendix C: Variants of the workflow for [2Fe-2S]

Table IV separates the contributions of the orbital loop and of the combination of the strings of all rounds, for noiseless SqDRIFT samples at $D = 10^6$. With a single orbital optimization per round and the strings of each round used alone, two rounds end at $\Delta E_\mathrm{P} = 0.31$ mE_h. The orbital loop lowers this to 0.28 mE_h, and adding the combination of the strings lowers it to 0.23 mE_h. Without the combination a third measurement does not help (0.29 against 0.28 mE_h), whereas with it ΔE_P falls to 0.21 mE_h, against 0.20 mE_h for CIPSI over all strings.

Appendix D: Sampled strings under orbital rotation

The orbital loop reuses strings sampled in one basis as occupation patterns of rotated orbitals. We tested this on [4Fe-4S] with samples measured on `ibm_kingston` in a fixed basis B'. This basis holds the orbitals after one quasi-Newton optimization of a first-round state. From 2×10^7 LUCJ shots and 1.4×10^5 SqDRIFT circuits measured in B', we followed two alternations of the orbital loop at $D = 10^7$ and $\epsilon = 5\times10^{-7}\,E_\mathrm{h}$ (Table V). Each orbital keeps more than 87% of its weight on the corresponding orbital of B', and the best one-to-one match of old and new orbitals is the identity, so the orbitals stay on the same sites. The mixing occurs mainly among the partly occupied orbitals. After two alternations the 20 leading determinants keep 77–84% of their weight on the same occupation patterns. Over the same alternations the share f of Eq. (3) stays at 0.97 (LUCJ) and 0.66 (SqDRIFT) to within 0.005, while ΔE falls from 15.59 to 13.43 mE_h (LUCJ) and from 14.86 to 13.17 mE_h (SqDRIFT). In this test the rotation does not make the sampled strings less useful. In the first orbital loops of the hardware rounds the orbitals rotate much further (Sec. IV). There f was

TABLE V. [4Fe-4S], strings sampled in a fixed basis B' at $D = 10^7$. The rows give ΔE (mE_h), the share f of the second-order correction inside $\mathcal{S}_\uparrow \times \mathcal{S}_\downarrow$, and the mean weight p that the 20 leading determinants keep on their occupation patterns of B', in B' and after one and two alternations of the orbital loop.

	B'	1 alternation	2 alternations
LUCJ, ΔE	15.59	14.18	13.43
LUCJ, f	0.972	0.972	0.971
LUCJ, p	1	0.91	0.77
SqDRIFT, ΔE	14.86	13.67	13.17
SqDRIFT, f	0.657	0.661	0.661
SqDRIFT, p	1	0.94	0.84

computed only at the end of each loop, so this test does not show how f changes under such large rotations.

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